

TIMOFEI ASINSKI

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EDUCATION

University of California, Santa Cruz

PhD in Computer Science and Engineering

Advised by Prof. Alexandra Kolla

Santa Cruz, CA

2026 – 2031 (Expected)

Northwestern University

B.S. in Computer Science and Mathematics, McCormick School of Engineering

Evanston, IL

2022 – 2026

Relevant Coursework: Approximation Algorithms, Real Analysis, Abstract Algebra, Probability Theory

ONGOING RESEARCH

- Asinski, T. *The Price of Robustness in Learned BSTs* (In preparation)
- Asinski, T., Makarychev, K., Shein, A. *Worst-Case and Average-Case Analysis of Byte Pair Encoding Tokenization* (In preparation)

TEACHING

Northwestern CS Department

Undergraduate Peer Mentor

Sep 2025 – March 2026

Evanston, IL

- CS 326: Introduction to the Data Science Pipeline
- CS 336: Design and Analysis of Algorithms

Hosted weekly office hours, graded assignments, and developed course materials and rubrics.

EXPERIENCE

Northwestern University CS Theory Group

Research Assistant – Algorithms

Jan 2025 – Present

Evanston, IL

Pursued two independent research projects under Dr. Konstantin Makarychev.

- Engineered efficient Python and C++ implementations of the Byte Pair Encoding (BPE) algorithm, and a brute-force optimal compression algorithm to examine BPE's empirical performance.
- Extended theoretical research on BPE tokenization, proving near-tight bounds on the worst-case performance of the algorithm.
- Answered an open question in learning-augmented algorithms research by creating novel learning-augmented binary search trees with strong worst-case performance guarantees.

FlowerAI

Software Engineer – Backend (Remote)

Oct 2024 – Feb 2025

Evanston, IL

Developed backend software for a cryptocurrency trading group.

- Designed and implemented a high-throughput cryptocurrency data processing and storage pipeline.
- Implemented a blockchain tracker ingesting 1,200+ Solana transactions daily via gRPC and PostgreSQL for real-time analytics.
- Created an API for storage and querying of trading performance metrics.

Northwestern VPD Group

Research Assistant – Computer Vision & Machine Learning

May 2023 – Sep 2024

Evanston, IL

Implemented computer vision tools for an electron microscopy lab.

- Developed a neural network identifying material composition from microscopy scans.
- Accelerated image processing scripts 10× using Python multiprocessing.
- Delivered reusable ML scripts for atomic simulations and data augmentation, enabling lab-wide use.

TECHNICAL SKILLS

Languages: Python, C++, Javascript, Go (Golang), C, SQL, Bash
Frameworks and Libraries: NumPy, PyTorch, Matplotlib, Pandas, Gin, FastAPI, OpenCV
Tools and Technologies: Linux, Git, Github, gdb, Tableau, OpenRefine

HONORS & AWARDS

Northwestern Summer Undergraduate Research Grant	Summer 2023
• Won \$16,000 grant to write and produce an album.	
McCormick Dean’s List	Spring 2026, Fall 2025, Spring 2025, Winter 2025, Fall 2024, Winter 2024
National Merit Finalist	Spring 2022